Anomaly Detection for Anti-Money Laundering: A Comprehensive Literature Review (2020–2026)

Literature Survey – Internal Reference

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Abstract

This document provides a comprehensive survey of anomaly detection methods applied to Anti-Money Laundering (AML), covering research from 2020 to early 2026. We organize the literature along seven dimensions: (1) graph and network-based approaches, (2) deep learning methods, (3) traditional machine learning, (4) contextual and KYC-enhanced detection, (5) feature engineering, (6) datasets and benchmarks, and (7) emerging paradigms including federated learning, explainability, and agentic AI. For each method category, we summarize key papers, techniques, datasets used, reported performance metrics, and practical considerations for deployment. This review targets a data scientist audience with prior anomaly detection experience who is beginning an AML-specific project.

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1 Introduction

Anti-Money Laundering (AML) is a critical compliance domain in financial services. Money laundering—the process of disguising illegally obtained funds as legitimate—involves three stages: *placement* (introducing illicit funds into the financial system), *layering* (obscuring the trail through complex transactions), and *integration* (re-entering cleaned funds into the economy).

Traditional AML systems rely heavily on rule-based transaction monitoring, which generates alerts when predefined thresholds are breached (e.g., transactions exceeding $10,000, rapid fund transfers across jurisdictions). These systems suffer from extremely high false positive rates—often 95–99% of generated alerts are not actionable—creating enormous operational costs for compliance teams.

Machine learning and deep learning approaches have emerged as powerful alternatives, with the potential to reduce false positives while maintaining or improving detection of genuinely suspicious activity. According to a 2025 SAS/KPMG/ACAMS survey, 62% of financial institutions already use AI/ML for AML in some capacity, though only 18% have fully production-deployed solutions.

This review is organized as follows: Section 2 covers graph and network-based approaches; Section 3 addresses deep learning methods; Section 4 covers traditional ML; Section 5 discusses contextual and KYC-enhanced detection; Section 6 reviews feature engineering; Section 7 surveys available datasets and evaluation methodology; and Section 8 covers emerging paradigms.

2 Graph and Network-Based Approaches

Financial transactions naturally form graphs: nodes represent accounts or entities, and edges represent fund transfers. Graph-based methods exploit this relational structure to detect money laundering patterns that are invisible in tabular representations. This section covers Graph Neural Networks (GNNs), heterogeneous graph methods, temporal graph approaches, and subgraph/community detection.

2.1 Graph Neural Networks (GNNs) for AML

GNNs have become a dominant paradigm for AML on transaction networks. A comprehensive 2024 survey by Cheng et al. ("Graph Neural Networks for Financial Fraud Detection: A Review," *Frontiers of Computer Science*, 2025) categorized over 100 GNN-based fraud detection methods, finding that GNNs substantially outperform traditional approaches by capturing relational patterns and dynamics within financial networks.

2.1.1 Foundational Work: Elliptic Dataset

Weber et al. (2019, KDD Workshop) introduced the **Elliptic dataset**—203,769 Bitcoin transactions with 234,355 directed edges—and evaluated GCN, Random Forest, and Logistic Regression. GCN achieved ∼0.97 AUC, demonstrating that graph structure improves over node features alone. This dataset remains the most widely used public AML benchmark.

2.1.2 Key GNN Architectures Applied to AML

- **GCN** (Graph Convolutional Networks): Baseline graph method; AUC ∼0.97, F1 (illicit) ∼0.52 on Elliptic.
- **GAT** (Graph Attention Networks): Attention-weighted neighbor aggregation; AUC ∼0.97, F1 ∼0.51 on Elliptic. Recent temporal-aware GAT (ATGAT) on Elliptic++ achieves AUC 0.913, a 9.2% improvement over XGBoost.

- **GraphSAGE**: Inductive (generalizable to unseen nodes); AUC ∼0.97, F1 ∼0.54 on Elliptic. F1 0.70–0.85 on AMLSim depending on laundering pattern complexity.
- **Improved HybridNet** (2025): Combines GraphSAGE convolutional layers with GATv2 attention and temporal encoding, capturing both local structure and dynamic patterns.
- **LineMVGNN** (2025): Line-graph-assisted multi-view GNN that considers both payment and receipt transaction views, addressing limitations of spectral GNNs on directed financial graphs.
- **LayerWeighted-GCN (LWG)**: Adaptive layer weighting for dynamic contribution of different GNN depths, improving capture of complex financial relationships.

2.1.3 Self-Supervised and Contrastive Learning on Graphs

Label scarcity is a fundamental challenge in AML (labeled SAR data is rare and sensitive). Self-supervised methods address this:

- **LaundroGraph** (Cardoso et al., 2022, ECML-PKDD Workshop): Self-supervised graph representation learning for AML; learns embeddings without labeled data, then applies downstream classifiers (typically XGBoost). XGBoost on graph features improved F1 by ∼5–10% over raw tabular features.
- **Bitcoin Money Laundering Detection via Subgraph Contrastive Learning** (BitCHetG, 2024): Contrastive learning on heterogeneous subgraphs for laundering group detection.
- **CONSISGAD** (Chen et al., 2024) and **POCL** (Zhang et al., 2024): Contrastive frameworks mitigating label imbalance, though confined to pairwise graphs.
- **HCLNet** (2025): Hypergraph contrastive learning combining hypergraph structure modeling with self-supervised contrastive learning for group-level fraud semantics.
- **Causal-DHG** (2024): Dynamic hypergraph framework integrating causal intervention and multi-view contrastive learning for robust fraud detection.
- **GRAD** (2025, TheWebConf): Guided relation diffusion for graph augmentation, using supervised graph contrastive learning to mine expert knowledge from limited labels.

2.2 Heterogeneous Graph Approaches

Real financial networks contain multiple entity types (persons, accounts, merchants, devices, IPs) and relation types (transfers, logins, ownership). Heterogeneous GNNs capture this richness:

- **CARE-GNN** (Dou et al., 2020, CIKM): Camouflage-Resistant GNN handling fraudsters who deliberately connect to benign nodes. Uses reinforcement learning-based relation selector. AUC ∼0.94 on YelpChi.
- **PC-GNN** (Liu et al., 2021, WWW): Pick-and-Choose neighbor sampling for class imbalance. AUC ∼0.97 on YelpChi.
- **xFraud** (Rao et al., 2022, VLDB): Industrial-scale heterogeneous GNN deployed at eBay-scale with explainability via attention visualization. Processes billions of nodes.
- **HAN** (Hierarchical Attention Network): Node-level and semantic-level (meta-path) attention; adapted for AML with 2–8% AUC improvement over homogeneous GNNs.
- **FRAUDRE** (Zhang et al., 2021): Dual-resistance framework handling both feature and relation camouflage.
- **MAHINDER**: Multi-view attributed heterogeneous information network model with hierarchical attention.
- **HTGNN** (2025): Heterogeneous Temporal Graph Neural Network integrating spatial, temporal, and semantic information for real-time fraud detection.

2.3 Temporal and Dynamic Graph Methods

Money laundering patterns evolve over time, and launderers continuously adapt tactics. Temporal graph methods are essential:

- **EvolveGCN** (Pareja et al., 2020, AAAI): Uses RNN (GRU/LSTM) to evolve GCN weight matrices over time. F1 (illicit) ∼0.58 on Elliptic temporal splits, improving over static GCN (∼0.52).
- **TGN** (Temporal Graph Networks, Rossi et al., 2020): Memory-augmented temporal graph network with per-node memory updated at each interaction. Highly relevant for transaction streams; adapted for AML by subsequent works.
- **Graph-Temporal Contrastive Transformer (GTCT)** (2025, Algorithms): Models both structural dependencies and temporal evolution in transactional behaviors.
- **Multi-Distance Spatial-Temporal GNN** (Chen, 2025): Multi-distance graph convolutional architecture capturing both local and global spatial dependencies in blockchain transactions.
- **Continual Graph Learning for AML** (Deprez et al., 2025, WIREs Computational Statistics): Comprehensive review of replay-based, regularization-based, and architecture-based continual learning strategies within GNN frameworks, addressing catastrophic forgetting.
- **DynGraphTransformer / Streaming GNNs**: Industry-academic collaborations (Ant Group, PayPal) developing streaming/online GNNs for real-time transaction processing on evolving graphs.

2.4 Subgraph and Community Detection for AML

Money laundering involves coordinated networks of accounts. Subgraph-level detection captures patterns invisible at the node level:

- **FlowScope** (Li et al., 2020, KDD): Detects dense subgraphs with flow patterns characteristic of money laundering (fan-out then fan-in / scatter-gather). Orders of magnitude faster than baseline subgraph enumeration.
- **SMoTeF** (2024, Applied Intelligence): Smurf money laundering detection using temporal order and flow analysis on time-evolving transaction networks. Applied to a real-world graph with 180M+ transactions across 31M accounts.
- **GARG-AML** (2025): Scalable and interpretable graph-based framework specifically for smurfing detection. Outperforms state-of-the-art smurfing detection methods on large-scale synthetic and open-source networks.
- **AMLGaurd** (2025): Unsupervised anomaly detection using Graph Auto-Encoders (GAEs) encoding structural and transactional information into latent space, identifying anomalies via reconstruction discrepancies.
- **Graph Feature Preprocessor** (2024, KDD): Library for detecting typical money laundering patterns in real-time, exploiting multicore parallelism and dynamic in-memory graphs for subgraph pattern mining on incoming transaction streams.
- **The Shape of Money Laundering** (2024): Subgraph representation learning on Elliptic2, treating Bitcoin addresses controlled by the same entity as nodes to reveal laundering scheme "shapes" (peeling chains, nested services).

2.5 Graph-Based AML: Summary Table

Table 1: Performance comparison of graph-based methods on standard benchmarks

Method	Dataset	AUC	F1 (illicit)	Key Note
Random Forest (baseline)	Elliptic	0.96	0.47	Node features only
GCN	Elliptic	0.97	0.52	+graph structure
GAT	Elliptic	0.97	0.51	Attention-based
GraphSAGE	Elliptic	0.97	0.54	Inductive learning
EvolveGCN-H	Elliptic	0.97	0.58	+temporal dynamics
ATGAT	Elliptic++	0.91	–	9.2% improvement over XGBoost
CARE-GNN	YelpChi	0.94	0.86	Anti-camouflage
PC-GNN	YelpChi	0.97	0.88	Imbalanced learning
GraphSAGE	AMLSim	0.85–0.95	0.70–0.85	Varies by pattern
GCN	AMLSim	0.80–0.92	0.65–0.80	Varies by pattern

3 Deep Learning Approaches (Non-Graph)

3.1 Sequence Models (RNN, LSTM, GRU)

Transaction sequences are naturally temporal, making recurrent architectures well-suited:

- **LSTM for AML on Mobile Transactions** (2025 survey by Kumari et al.): RNN-based methods model dependencies in sequential transaction data, with LSTM and GRU addressing vanishing gradient issues. GRU achieves the highest results with AUC-ROC above 0.7 on AML-specific data.
- **Attention-enhanced LSTM** (Niu et al., 2021, Journal of Big Data): Credit card fraud detection using attention mechanisms + LSTM. The attention mechanism allows the model to focus on the most suspicious parts of transaction sequences.
- **Jensen and Iosifidis (2023)**: Deep learning to replace predefined rules with automatically extracted characteristics from transaction sequences at the bank level. The core idea is automatic feature extraction from sequential transaction patterns.
- SimpleRNN models face significant challenges with long input sequences due to vanishing gradients, whereas LSTM and GRU demonstrate superior performance. GRU is often preferred for AML due to fewer parameters and comparable performance to LSTM.

3.2 Transformer-Based Models

Transformers have emerged as powerful alternatives to RNNs for sequential transaction modeling:

- **Tang and Liu (2024)**: Unified Transformer-BDI architecture using distributed knowledge distillation. F1-score 92.87%, accuracy 98.98%, AUC 96.73%.
- **Transformer-Based Behavioral Modeling** (2024–2025): Uses GPT-style pretraining on large-scale payment data to reconstruct behavioral patterns and detect anomalies in real-time.
- **TREASURE** (KDD 2026): Transformer-based foundation model for high-volume transaction understanding, representing the frontier of pretrained models for financial transactions.
- **TransactionGPT** (2025): Generative pretrained model for transaction sequence understanding.
- **FraudGT** (2024, ICAIF): Graph Transformer for financial fraud detection, combining graph structure with transformer attention.

- **Credit Card Fraud Detection Using Advanced Transformer Model** (2024): Multiple transformer variants (TabTransformer, TransTab, BERT, RoBERTa) applied to extract temporal and contextual features from structured transaction data.
- **Spatial-Temporal-Aware Graph Transformer** (2024, IEEE TII): Combines spatial graph structure with temporal transformer attention for transaction fraud detection.

3.3 Autoencoders and Variational Methods

Autoencoders are naturally suited for anomaly detection—trained on normal transactions, high reconstruction error signals anomalies:

- **VAE for Credit Card Anomaly Detection** (2024, Big Data Mining and Analytics): Evaluates convolutional network-based VAE models on credit card datasets. VAE-based unsupervised detection trained exclusively on legitimate transactions, with hybrid models combining VAE anomaly detection with classifiers for fraud labeling.
- **GAN+VAE Hybrid** (2025): Combines GANs and VAEs to detect abnormal behaviors in large payment flows. VAE models the latent distribution of payment flows to ensure generated data resembles real transaction features.
- **LVAESC-SFD** (2026): Correlation-based Feature Selection with Variational Autoencoder achieving 98.24% accuracy, outperforming existing models.
- **Denoising Autoencoder for AML** (Bundesbank, 2023): Explaining anomalies using denoising autoencoders for financial tabular data, combining detection with interpretability.
- **Self-Attention + VAE** (2024, Expert Systems with Applications): Fund transfer fraud detection combining self-attention mechanisms with autoencoders for long-range dependency identification in transaction sequences.
- Shallow autoencoders (1–2 hidden layers) are often sufficient for AML on tabular data; deep autoencoders show marginal improvement with significantly more complexity (AUC $\sim$0.85–0.90 on credit card fraud; $\sim$0.80–0.85 on AML-specific data).

3.4 Generative Models (GANs) for AML

GANs serve dual purposes in AML: synthetic data generation for training and direct anomaly detection:

- **Wasserstein GAN (WGAN) for AML Data Augmentation**: Generates synthetic fraudulent transactions to address extreme class imbalance. F1-scores improve by 10–15% with GAN-augmented training data.
- **SMOTE-GAN Hybrids** (2023, Finances): Explores combinations of SMOTE and GANs including SMOTified-GAN, SMOTE+GAN, and GANified-SMOTE for addressing class imbalance in fraud detection.
- **Advanced R-GAN** (2024): Regularized GAN generating anomaly data for improved detection in imbalanced datasets using specialized activation functions and spectral normalization to mitigate mode collapse.
- **CTGAN, Copula GAN, TVAE** (various, 2023–2025): State-of-the-art algorithms for synthesizing realistic AML training data. Synthetic transaction data achieves 96–99% utility equivalence to production data for AML model testing.
- **Utilizing GANs for Fraud Detection** (2024): GAN-generated fraudulent transaction data used to train fraud-detection systems, teaching the system to distinguish between suspicious and genuine transactions.

3.5 LLMs and Foundation Models for AML

An emerging frontier with rapid development in 2024–2026:

- **FLAG** (KDD 2025): Fraud Detection with LLM-enhanced GNN, combining language model reasoning with graph neural network structure.
- **GuARD** (KDD 2025): Effective anomaly detection through text-rich and graph-informed language model.
- **DGP** (AAAI 2026): Dual-Granularity Prompting Framework for fraud detection with graph-enhanced LLMs.
- **AuditCopilot** (NeurIPS 2025 Workshop): Leveraging LLMs for fraud detection in double-entry bookkeeping.
- **Co-Investigator AI** (2025): Agentic framework optimized to produce Suspicious Activity Reports (SARs) significantly faster and with greater accuracy than traditional methods.
- **PANTHER** (NeurIPS 2025): Generative pretraining beyond language for sequential user behavior modeling.
- **FinGPT** (IJCAI 2023): Open-source financial large language model, with downstream applications in compliance and AML narrative generation.

4 Traditional Machine Learning Approaches

4.1 Gradient Boosting Methods

Gradient boosting remains the dominant paradigm for tabular AML data, consistent with the broader finding that tree-based models outperform deep learning on structured data (Grinsztajn et al., 2022, NeurIPS).

- **XGBoost**: Most frequently used supervised classifier in AML studies. AUC 0.95–0.99 across multiple datasets. The IEEE-CIS Fraud Detection competition (2019–2020) saw all top-10 solutions use XGBoost or LightGBM, with winning AUC $\sim$0.966.
- **LightGBM**: Within 0.5% AUC of XGBoost but 3–5$\times$ faster training. Histogram-based splitting and leaf-wise growth particularly effective for high-cardinality categorical features. Practical advantage for production AML systems requiring frequent retraining.
- **CatBoost**: Native categorical feature handling with ordered boosting to reduce target leakage. AUC $\sim$0.97–0.98. Advantage over XGBoost/LightGBM disappears when manual target encoding is applied.
- **Stacking Ensembles**: RF + XGBoost + LightGBM with logistic regression meta-learner. Typically improves AUC by 0.5–1.5% over best single model. Used in production at several banks.

A recent stacking framework combining XGBoost, LightGBM, and CatBoost with explainable AI achieved 99% accuracy, 0.99 precision, recall, F1, and AUC on large-scale real-world data (2025).

4.2 Unsupervised Anomaly Detection

Particularly valuable when labeled SAR data is scarce or unavailable:

- **Isolation Forest**: Best general-purpose unsupervised anomaly detector for AML. Low computational complexity, high scalability, robust to noise and irrelevant features. AUC $\sim$0.78–0.85. Most effective as feature extractor ("anomaly score as feature") rather than standalone detector—two-stage IF scores + RF improves AUC by $\sim$3% over RF alone (Jullum et al., 2020).
- **Extended Isolation Forest (EIF)**: Non-axis-parallel splits addressing "banding" artifact. Better calibrated anomaly scores; $\sim$1–2% AUC improvement over standard IF. Preferred for AML due to better handling of correlated features.
- **Local Outlier Factor (LOF)**: Captures local density deviations. AUC $\sim$0.72–0.80. Better for small, low-dimensional subsets but doesn't scale well.

- **One-Class SVM**: Trained only on legitimate transactions; no labeled fraud required. AUC 0.70–0.80 standalone. Useful as complementary signal in ensembles.
- **DBSCAN**: Identifies structural outliers in transaction embeddings. Recall ∼0.65, Precision ∼0.15 standalone—best as pre-filter. Useful for multi-stage smurfing/layering chain detection.
- **PCA Reconstruction**: Fast and interpretable. AUC ∼0.70–0.85 on credit card fraud; lower (∼0.70–0.75) on AML-specific data. Robust PCA decomposes data into low-rank (normal) + sparse (anomaly) components.

Comparative ranking (Hilal et al., 2022): IF > Autoencoder > LOF > OCSVM > DBSCAN for AML-type detection.

4.3 Rule-Based vs. ML Hybrid Architectures

Pure ML replacement of rules is rare due to regulatory requirements. The dominant industry pattern is a staged architecture:

1. **Hard rules** for regulatory compliance (e.g., CTR thresholds)
2. **Soft rules** for known typologies (structuring, round-tripping)
3. **ML models** for emerging/unknown patterns

Key findings:

- Rocha-Salazar et al. (2021): Hybrid approach where rules generate candidate alerts and RF re-scores/prioritizes. Alert volume reduced 70%; SAR precision improved from 2% (rules-only) to 15% (hybrid).
- Soltani Delgosha and Hajiheydari (2021): Staged rule+ML architecture reduced false positives by ∼40% versus rules-only while maintaining recall.
- **FRAML Framework** (2023): Three-tier production architecture satisfying both detection efficacy and regulatory explainability requirements.
- Regulators (FinCEN, FCA, EBA) mandate explainable triggers, driving hybrid architectures where rules provide audit trail and ML provides ranking.

4.4 Traditional ML: Performance Summary

Table 2: Performance comparison of traditional ML methods for AML

Method	Typical AUC	Label Req.	Strengths	Weaknesses
XGBoost	0.95–0.99	Supervised	Best tabular perf.; SHAP support	Requires tuning
LightGBM	0.95–0.98	Supervised	Fastest training; large-scale	Slightly lower on small data
CatBoost	0.95–0.98	Supervised	Native categoricals	Less community adoption
Random Forest	0.90–0.96	Supervised	Robust; less tuning	Lower peak performance
Stacking	0.96–0.99	Supervised	Best combined	Complexity; slower inference
Isolation Forest	0.78–0.85	Unsupervised	No labels; fast; scalable	Lower precision standalone
LOF	0.72–0.80	Unsupervised	Local anomalies	Doesn't scale
One-Class SVM	0.70–0.80	Unsupervised	Only needs normal class	Kernel choice critical
PCA	0.70–0.85	Unsupervised	Fast; interpretable	Misses nonlinear patterns
DBSCAN	0.60–0.75	Unsupervised	Structural outliers	Low precision

5 Contextual and KYC-Enhanced Detection

5.1 KYC Data Integration

Contextual anomaly detection conditions suspicious behavior thresholds on customer profiles. The key insight is that a $50,000 wire transfer is normal for a corporate account but anomalous for a student account:

- **Customer 360 Profiles**: Dynamic integration of KYC data, transaction details, and external datasets enabling trend identification, anomaly detection, and peer group comparisons.
- **Dynamic Risk Assessment**: HSBC deployed ML models analyzing transaction patterns, network behaviors, and KYC data, achieving a **60% reduction in alerts** and a **2–4× increase in true positive detection**.
- **Risk Rating Models**: AI-driven customer risk ratings incorporating behavioral and transaction-based factors, sharpening accuracy in transaction monitoring.
- Customer segmentation by behavioral profiles (k-means, DBSCAN) enables contextual thresholding: customers far from their cluster centroid receive heightened scrutiny.

5.2 Entity Resolution and Identity Matching

Laundering networks exploit fragmented identities across systems:

- **Knowledge Graphs for AML**: Connect people, companies, accounts, transactions into searchable networks. Trace ownership chains across borders and shell entities. Link multiple records of the same customer across systems.
- **Data-driven name matching**: 90% more accuracy than legacy fuzzy matching, handling transliterations, scripts, and real-world name variants.
- **Graph-based entity resolution** using Neo4j and similar technologies for uncovering non-obvious relationships (shared addresses, shared phones, shared IPs).
- Composite AI approaches combining entity resolution with graph analytics enable proactive discovery rather than reactive alert-handling.

5.3 Behavioral Profiling and Contextual Features

Key behavioral features for contextual detection:

- Ratio of transaction amount to customer's historical mean/median
- Z-score of transaction relative to customer history
- New merchant/country/channel flags (first-time behavior)
- Dormancy features (suddenly active account)
- Day-of-week and time-of-day deviation from usual patterns
- Recency of KYC update; days since last address change
- Country risk scores (FATF greylist/blacklist)
- PEP (Politically Exposed Person) and sanctions screening flags

6 Feature Engineering for AML

Feature engineering matters more than model choice for AML. The gap between a well-engineered XGBoost and a poorly-engineered deep model is much larger than between model architectures.

6.1 Feature Categories

Table 3: Taxonomy of AML features

Category	Examples	Impact
Velocity	Txn count/amount in last 1h, 6h, 24h, 7d, 30d; unique counterparty count; frequency acceleration; time since last txn	AUC improvement ∼5–8% (Bahnsen et al., 2016)
Behavioral	Amount ratio to historical mean; Z-score; new merchant/country flag; dormancy; day/time deviation	Single most important category for AML
Network/Graph	In/out-degree; PageRank; clustering coefficient; fan-in/fan-out; cyclic detection; community membership	F1 improvement ∼7% even without GNNs (Alarab et al., 2020)
Aggregation	Per-customer stats (mean, std, skew); per-merchant fraud rate; per-country-pair transfer amounts; per-channel statistics	Foundation for behavioral deviation
Temporal	Day of week; hour; is_weekend; is_holiday; days since account opening; seasonal patterns	Captures laundering timing patterns
Risk Indicators	FATF country risk; customer risk rating; PEP flag; industry risk; historical SAR count	Contextual conditioning

6.2 Two-Stage Feature Architecture

A trend gaining popularity (2023–2025) is the two-stage approach:

1. **Stage 1**: Unsupervised scoring (Isolation Forest anomaly scores, PCA reconstruction errors, autoencoder reconstruction loss, cluster distances)
2. **Stage 2**: Supervised classification on enriched features (original features + Stage 1 scores)

This addresses label scarcity: unsupervised signals provide additional discriminative power even when labeled SAR data is limited. Jullum et al. (2020) demonstrated ∼3% AUC improvement using this paradigm.

7 Datasets, Benchmarks, and Evaluation

7.1 Public Datasets

Table 4: Public datasets for AML research

Dataset	Type	Size	Imbalance	Notes
Elliptic	Real BTC	203K txns, 234K edges	2% illicit	Most-used AML benchmark; 49 time steps; 166 features
Elliptic++ (2023–2024)	Real BTC	204K txns + address-level	2% illicit	Extends Elliptic with actor-level graph; subgraph-level labels
AMLSim (IBM)	Synthetic	Configurable	Configurable	Open-source; controllable laundering patterns (fan-out, fan-in, cycle, scatter-gather)
PaySim	Synthetic	6.3M txns	0.13% fraud	Agent-based mobile money simulation; network structure
IBM AML Synthetic	Synthetic	180M+ txns	Configurable	Large-scale multi-bank; multiple laundering typologies
IEEE-CIS Fraud	Real CC	590K txns	3.5% fraud	Kaggle competition; rich features
European CC Fraud	Real CC	284K txns	0.17% fraud	PCA-anonymized; very imbalanced
Bitcoin-OTC/Alpha	Real BTC	5.8K/3.7K nodes	–	Trust-weighted signed networks; less directly AML
YelpChi / Amazon	Reviews	67K/11K reviews	–	Review fraud; standard heterogeneous GNN benchmarks

Synthetic Data Generation. A 2023 Nature Scientific Data paper by Altman et al. (IBM Research) introduced a large-scale synthetic dataset benchmarking AML methods at NeurIPS 2023. The synthetic data generation market is projected to grow from $310–576M (2024) to $1.8–16.7B by 2030, driven by AML/fraud applications. By 2025, 75% of large banks rely on synthetic data for AI projects.

Key synthetic data tools: CTGAN, Copula GAN, Gaussian Copulas, TVAE—achieving 96–99% utility equivalence to production data for AML model testing.

Alan Turing Institute Project. A UK initiative (commenced autumn 2024) using GANs to create fully synthetic AML datasets for cross-institution model development without privacy violations.

7.2 Class Imbalance Handling

AML datasets exhibit extreme imbalance (0.1–2% positive). Key approaches:

- **SMOTE**: Most widely used oversampling; RF+SMOTE improved F1 from 0.72 to 0.87 (Al-Hashedi and Magalingam, 2021). SMOTE+RF outperforms ADASYN+RF.
- **SMOTE-GAN Hybrids**: Combining statistical and generative oversampling for enhanced class balance.
- **Cost-sensitive learning**: Class weights in loss functions; focal loss adaptations.
- **GraphSMOTE**: Graph-aware oversampling for GNN-based methods.
- **PC-GNN sampling**: Pick-and-choose neighbor sampling (Liu et al., 2021).
- **AMLNet** (2025): Combines SMOTE and RandomUnderSampler with multi-agent knowledge-based framework.

7.3 Evaluation Metrics

Standard academic metrics (AUC, F1) are necessary but insufficient for AML:

- **AUC-ROC**: Standard ranking metric. Most methods achieve 0.95+ on Elliptic, making fine-grained comparison difficult.
- **F1 (illicit class)**: More discriminating than AUC on imbalanced data. Ranges from 0.47 (RF baseline) to 0.88 (PC-GNN on YelpChi).
- **Precision at top-k**: Operational metric—how many of the top-k alerts are true positives? Directly maps to analyst workload.
- **Alert-level precision**: Percentage of generated alerts that result in SAR filings. Rule-based systems: ∼2%; hybrid ML: ∼15%.
- **False positive reduction**: Primary industry KPI. ML systems consistently achieve 40–60% FP reduction over rules.
- **Population Stability Index (PSI)**: Monitors concept drift; triggers model retraining.

7.4 Concept Drift and Model Maintenance

Money launderers continuously adapt tactics, causing concept drift:

- **Continual Graph Learning for AML** (Deprez et al., 2025): Reviews replay-based, regularization-based, and architecture-based strategies to prevent catastrophic forgetting when updating AML models.
- **Cross-Regional Continual Learning** (TKDE 2024): Knowledge transfer across geographic regions for AML with distribution shift.
- Traditional models suffer from catastrophic forgetting when fine-tuned on new data, limiting effectiveness in dynamic environments.
- PSI-based drift detection commonly used to trigger retraining cycles in production.

8 Emerging Paradigms

8.1 Federated Learning for Cross-Institution AML

Banks cannot share customer data, but laundering networks span institutions. Federated learning enables collaborative model training without data sharing:

- **FAML Framework**: Federated Anti-Money Laundering enabling multiple banks to collaboratively train models without compromising data privacy or regulatory compliance (GDPR, etc.).
- **FedGraph-VASP** (2026): Privacy-preserving federated graph learning with Boundary Embedding Exchange protocol secured by Post-Quantum Cryptography. Enables cross-institutional GNN training for AML.
- Federated AML systems achieve up to **30% higher accuracy** versus isolated approaches. One implementation on PaySim reported 0.999 accuracy with 0.1% false alarm rate.
- **IBM Research**: Building privacy-preserving federated learning specifically for financial crime detection.
- **Fully Homomorphic Encryption (FHE) + GNNs**: Privacy-preserving graph-based ML for collaborative AML without revealing raw data.
- **Federated Graph Anomaly Detection via Disentangled Representation Learning** (TheWebConf 2025): Disentangled representations enabling federated anomaly detection across institutions.
- **NVIDIA Federated Learning Application Runtime (NVFLARE)**: Platform used for deploying federated AML models at scale.

8.2 Explainability and Regulatory Compliance

Regulatory bodies (FinCEN, FCA, EBA, FATF) increasingly require model explainability:

- **SHAP/TreeSHAP**: De facto standard for AML explainability. Per-alert SHAP explanations improve analyst efficiency by ∼30% (Jullum et al., 2020). "Payment Format" identified as most crucial feature in DNN-based AML (recall 0.94).
- **LIME**: Less stable than SHAP on tabular data with correlated features (common in AML). Useful for quick per-alert explanation but SHAP generally preferred.
- **GNNExplainer adaptations**: Subgraph and attention-based explanations for graph models.
- **SEFraud** (KDD 2024): Self-explainable fraud detection via interpretative mask learning.
- **Counterfactual explanations**: "This alert would not have fired if..." format demanded by regulators.
- **Global surrogate models**: Interpretable decision trees/GAMs approximating black-box AML models for regulatory documentation.
- **Regulatory requirements**: EU AML Directive (2020), FinCEN guidance (2021) explicitly mandate "understandable" automated detection. The top barrier to AI adoption in AML shifted from budget (previously #1) to lack of regulatory imperative (37%, 2025 survey).

8.3 Agentic AI for AML Compliance

The most recent frontier (2025–2026), representing a paradigm shift in AML operations:

- **Nasdaq Verafin Agentic AI Workforce** (2025): Suite of digital workers automating low-value, high-volume AML compliance processes. Enables automation of up to 50% of AML workloads.
- **Co-Investigator AI** (2025): Agentic framework for producing SARs faster and more accurately than traditional methods.
- **McKinsey Report** (2025): Agentic AI can automate EDD review tasks including adverse media search, historic activity analysis, and case escalation decisions.
- **Agentic KYC workflows**: Agent-based architectures cut onboarding cycle times from weeks to hours while preserving auditability.
- Human-in-the-loop design: AI handles lower-risk tasks; humans retain high-level investigative and policy decisions.

8.4 Adversarial Robustness

Laundering networks actively adapt to evade detection:

- **Adversarial attacks on GNNs**: Topology perturbation, feature manipulation targeting graph-based detectors.
- **CARE-GNN** addresses camouflage where fraudsters deliberately connect to benign nodes.
- **RoBCtrl** (2025): Attacking GNN-based bot detectors via reinforced manipulation, highlighting vulnerability of deployed systems.
- **Adversarially Robust Temporal Graph Contrastive Learning** (2025): Defensive techniques for temporal graph AML models.
- **Graph Adversarial Training** (2024): Improving robustness of GNN-based anomaly detection through adversarial training regimes.

9 Taxonomy and Decision Framework

Table 5: Decision matrix: choosing an approach based on data availability and constraints

Scenario	Data Type	Labels	Recommended	Rationale
Tabular transactions, labeled SARs available	Tabular	Yes	XGBoost/LightGBM + SHAP	Highest performance on tabular; explainable
Tabular, no labels	Tabular	No	Isolation Forest + two-stage	IF scores as features for future supervised model
Transaction network available, labeled	Graph	Yes	GraphSAGE / GAT + temporal	Exploits relational structure
Transaction network, no labels	Graph	No	AMLGaurd (GAE) or self-supervised GNN	Unsupervised graph anomaly detection
Multi-institution collaboration	Any	Mixed	Federated learning (FAML)	Privacy-preserving cross-institution
Real-time monitoring	Streaming	Any	TGN / Streaming GNN	Handles continuous transaction streams
Regulatory compliance focus	Any	Any	Hybrid rule+ML + SHAP	Satisfies explainability requirements
Limited labeled data	Any	Few	Self-supervised (Laundro-Graph) + fine-tuning	Maximizes limited label value

10 Industry Adoption and Deployment

Table 6: Notable industry deployments of ML-based AML systems

Organization	Method	Scale	Year
HSBC	Dynamic Risk Assessment (ML+KYC)	Banking-scale	2024
Ant Group (Alipay)	Heterogeneous GNN	Billions of nodes	2020+
eBay	xFraud (Het-GNN)	Billions of transactions	2022
PayPal	GNN + streaming	Real-time fraud scoring	2021+
Feedzai	Graph-enhanced ML	Banking AML	2022+
HSBC + Google Cloud	GNN for AML	Banking-scale	2023
Elliptic	AI on Bitcoin	200M transactions	2024
Nasdaq Verafin	Agentic AI Workforce	Banking AML	2025

Adoption survey (SAS/KPMG/ACAMS, 2025):

- 18% have AI/ML in production; 18% piloting; 25% plan to implement in 12–18 months; 40% no current plans.
- Top priority: false positive reduction (38%, up 8% since 2021).
- Machine learning was the top technology choice at 58% (up 6% since 2021).
- Top barrier shifted from budget to "lack of regulatory imperative" (37%).
- 51% say regulators promote AI innovation (down 15 points from 2021); regulatory caution rising.

11 Open Challenges and Future Directions

1. **Extreme class imbalance**: Illicit transactions are 0.1–2% of volume. Even sophisticated sampling and cost-sensitive methods struggle with the rarest, most sophisticated laundering schemes.

2. **Concept drift and adversarial adaptation**: Launderers actively adapt to evade detection. Continual learning methods are nascent; most production systems rely on periodic retraining rather than online adaptation.

3. **Label noise and partial labels**: SAR labels are noisy (filed based on suspicion, not confirmed criminality) and partial (many illicit transactions go undetected). This fundamentally limits supervised approaches.

4. **Cross-institution collaboration**: Money laundering spans multiple institutions. Federated and privacy-preserving approaches are promising but face practical challenges (graph structure across institutions, regulatory fragmentation).

5. **Explainability gap**: Regulators demand explainability, but the most powerful models (deep GNNs, transformers) are the least interpretable. Bridging this gap remains critical for deployment.

6. **Real-time processing**: Most academic methods are batch-oriented. Production AML requires real-time or near-real-time scoring on millions of daily transactions.

7. **Cryptocurrency and DeFi**: New asset classes create new laundering vectors. The regulatory framework (FinCEN proposed rule June 2024) is still catching up.

8. **Graph foundation models**: Pre-trained graph models transferable across institutions/datasets are an emerging frontier (AnomalyGFM, KDD 2025), but early stage.

9. **Evaluation methodology**: Over-reliance on AUC on small public datasets (Elliptic) may not reflect real-world deployment performance. Need for better operational metrics and benchmarks.

10. **Data quality and integration**: Real AML systems must integrate heterogeneous data sources (core banking, SWIFT messages, KYC records, adverse media, sanctions lists) with varying quality and timeliness.

12 Conclusion

The AML anomaly detection landscape has undergone significant evolution from 2020 to 2026. Key trends include:

- **Graph methods have matured** from basic GCN to temporal, heterogeneous, and continual graph learning, with industrial deployments at major financial institutions.

- **Gradient boosting (XGBoost/LightGBM) remains king on tabular data**, and feature engineering matters more than model architecture for structured transaction data.
- **Hybrid rule+ML architectures** are the practical industry standard, balancing detection power with regulatory compliance.
- **Self-supervised and contrastive learning** address the fundamental challenge of label scarcity in AML.
- **Transformers and LLMs** are the newest frontier, with foundation models for transactions (TREASURE, TransactionGPT) and LLM-enhanced graph methods (FLAG, GuARD) emerging in 2025–2026.
- **Federated learning and privacy-preserving methods** enable cross-institutional collaboration without data sharing, with early results showing 30% accuracy improvements.
- **Agentic AI** represents the operational frontier, automating up to 50% of AML workloads including SAR generation and EDD reviews.
- **Explainability (SHAP)** is a non-negotiable requirement for production deployment.

For a practitioner starting an AML anomaly detection project, the recommended path is: begin with well-engineered features + gradient boosting as a strong baseline; add graph features (even without GNNs); evaluate GNN approaches if transaction network data is available; implement SHAP for explainability; and consider federated approaches for cross-institution scenarios.

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Note: This literature review was compiled in February 2026. Some citation details (exact page numbers, volume numbers) should be verified against original sources before inclusion in a formal publication. Performance metrics are reported as found in the literature and may reflect specific experimental configurations.